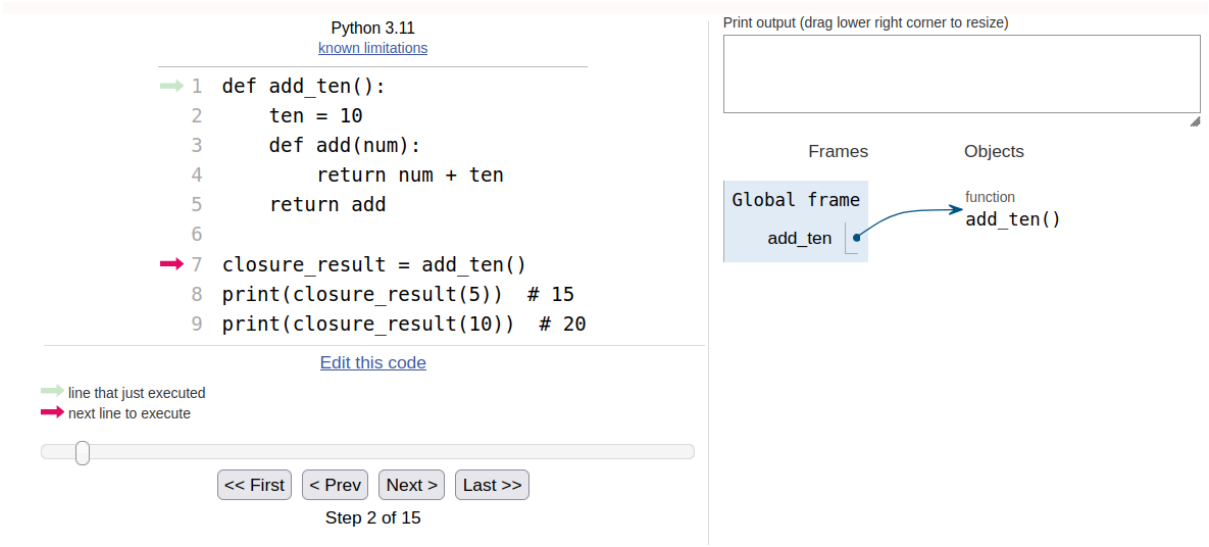
PYTHON CLOSURES.

Python permite que una función anidada acceda al ámbito externo de la función de cierre. Esto se conoce como cierre. Echemos un vistazo a cómo funcionan los cierres en Python. En Python, el cierre se crea anidando una función dentro de otra función de encapsulación y luego devolviendo la función interna. Vea el ejemplo a continuación.

```python
# CÓDIGO EJEMPLO
def add_ten():
    ten = 10
    def add(num):
        return num + ten
    return add

closure_result = add_ten()
print(closure_result(5))  # 15
print(closure_result(10))  # 20
```



Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

Edit this code

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next line to execute

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Step 3 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()

add_ten

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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Step 5 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()

add_ten
ten 10

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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Step 6 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()

f1: add_ten
add → function add(num) [parent=f1]
ten 10

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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Step 7 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()
closure_result

f1: add_ten
add → function add(num) [parent=f1]
ten 10
Return value

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

Edit this code

line that just executed
next line to execute

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Step 8 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()
closure_result → function add(num) [parent=f1]

f1: add_ten
add
ten 10
Return value

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

Edit this code

line that just executed
next line to execute

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Step 9 of 15

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()
closure_result → function add(num) [parent=f1]

f1: add_ten
add
ten 10
Return value

add [parent=f1]

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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next line to execute

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Step 10 of 15

NEW: teachers get free access to ad-free/AI-free mode

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()
closure_result → function add(num) [parent=f1]

f1: add_ten
add
ten 10
Return value

add [parent=f1]
num 5

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

Edit this code

line that just executed
next line to execute

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Step 11 of 15

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

Frames | Objects

Global frame
add_ten → function add_ten()
closure_result → function add(num) [parent=f1]

f1: add_ten
add
ten 10
Return value

add [parent=f1]
num 5
Return value 15

Python 3.11
known limitations

```
1  def add_ten():
2      ten = 10
3      def add(num):
4          return num + ten
5      return add
6
7  closure_result = add_ten()
8  print(closure_result(5))  # 15
9  print(closure_result(10))  # 20
```

Edit this code

line that just executed
next line to execute

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Step 12 of 15

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

```
15
```

Frames | Objects

Global frame
- add_ten → function: add_ten()
- closure_result → function: add(num) [parent=f1]

f1: add_ten
- add
- ten: 10
- Return value

Python 3.11
known limitations

```
1  def add_ten():
2      ten = 10
3      def add(num):
4          return num + ten
5      return add
6
7  closure_result = add_ten()
8  print(closure_result(5))  # 15
9  print(closure_result(10))  # 20
```

Edit this code

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next line to execute

<< First | < Prev | Next > | Last >>

Step 13 of 15

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

```
15
```

Frames | Objects

Global frame
- add_ten → function: add_ten()
- closure_result → function: add(num) [parent=f1]

f1: add_ten
- add
- ten: 10
- Return value

add [parent=f1]
- num: 10

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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next line to execute

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Step 14 of 15

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

```
15
```

Frames | Objects

Global frame
- add_ten → function add_ten()
- closure_result → function add(num) [parent=f1]

f1: add_ten
- add
- ten: 10
- Return value

add [parent=f1]
- num: 10

Python 3.11
known limitations

```
1 def add_ten():
2     ten = 10
3     def add(num):
4         return num + ten
5     return add
6
7 closure_result = add_ten()
8 print(closure_result(5))  # 15
9 print(closure_result(10))  # 20
```

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Step 15 of 15

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

```
15
```

Frames | Objects

Global frame
- add_ten → function add_ten()
- closure_result → function add(num) [parent=f1]

f1: add_ten
- add
- ten: 10
- Return value

add [parent=f1]
- num: 10
- Return value: 20

Python 3.11
known limitations

```
1  def add_ten():
2      ten = 10
3      def add(num):
4          return num + ten
5      return add
6
7  closure_result = add_ten()
8  print(closure_result(5))  # 15
9  print(closure_result(10))  # 20
```

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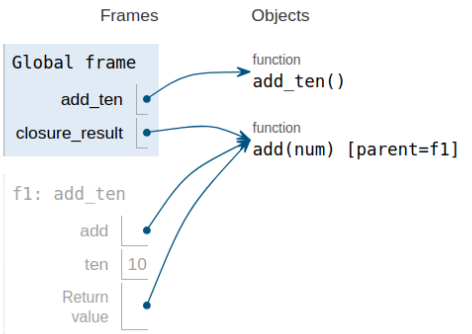
Done running (15 steps)

NEW: teachers get free access to ad-free/AI-free mode

Move and hide objects (instructions)

Print output (drag lower right corner to resize)

```
15
20
```